

Unified Consciousness Theory I v1.1

Ontic Structure, Physical Views, and the Structural Continuity from Physical Process to Conceptual Self

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Versioning statement. This manuscript is a successor draft to *Unified Consciousness Theory I: From Experience Existence to Experiential Structure*, v1.0. It does not overwrite or retroactively alter v1.0. The main purposes of v1.1 are to (i) separate ontic process structure from scientific views and estimates, (ii) state explicitly the strength of the structural-identity commitment used by later derivations, (iii) repair theorem dependencies, (iv) formalize the distinction between metaphysical identity and finite empirical bridge tests, and (v) develop a structural-evolutionary synthesis from elementary physical processes to human conceptual selfhood without treating evolution as a proof of the foundational axioms.

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Abstract

Most theories of consciousness search for a physical or computational property that separates conscious from nonconscious systems. Unified Consciousness Theory (UCT) begins from a different ontological commitment: every actual valid physical process token possesses non-empty experiential structure. Under this assumption, the universal existence question is no longer the principal discriminative problem. The central problem becomes the organization of experience.

Version 1.1 sharpens the structural identity thesis introduced in UCT I v1.0. It distinguishes an actual process token P , its complete token-relative ontic organization $\mathbf{D}^{ontic}(P)$, a finite scientific view $D_v(P)$, an estimated model $\widehat{D}_v(P)$, and any later subject-individuation scheme. The strengthened development-branch identity commitment, A5-OI, states that for actual process tokens within a declared common structural category, experiential and ontic physical structures share the same structural isomorphism classes:

$$\Phi(P) \cong \Phi(Q) \iff \mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q).$$

This is stronger than the one-way phenomenal-completeness formula explicitly stated in v1.0 and is therefore treated as a versioned strengthening rather than a retroactive reinterpretation.

The clarified architecture yields several conditional results. Under A1, no nonuniversal threshold of integration, recurrence, global access, explicit memory, higher-order representation, complexity, or related organization can equal the universal experience-existence predicate. Under the one-way completeness component of A5, present phenomenology is invariant to merely external probe choices, future facts not currently instantiated, and

causally erased history. Under A5-OI, nonfactorizable macroprocess structure entails non-product experiential structure, while microphysical changes may coexist with preserved macroexperience when the actual macroprocess organization is unchanged. A symmetry-based theorem shows that overlapping, structurally equivalent process candidates do not in general determine one unique mutually exclusive subject partition without additional symmetry-breaking structure.

The paper then develops an evolutionary-structural synthesis. Elementary interaction, self-maintaining cellular organization, memory traces, multicellular coordination, neural routing, affective organization, relational cognition, self-models, and the conceptual human “I” are treated as transformations of process organization rather than successive creations of a new phenomenal substance. Comparative biological findings—from non-neural habituation and cnidarian associative learning to insect relational abstraction, cephalopod episodic-like memory and delayed control, and the evolutionary layering of nociception and pain—are used only as illustrative structural anchors; they do not prove A1 or A5.

Finally, v1.1 separates the metaphysical identity commitment from its scientific descendants. Complete $\mathbf{D}^{ontic}$ and complete Φ are not ordinarily observed directly. Finite empirical tests therefore require a preregistered physical view, a bridge-sufficiency hypothesis, a declared phenomenal target, and explicit measurement assumptions. Failure at a frozen token/view remains failure of that operational bridge version; post-hoc token or scale changes constitute a new hypothesis. Such a failure does not, by itself, isolate A5-OI from all auxiliary bridge and measurement assumptions. UCT is therefore presented as a metaphysical-formal core with explicit, failure-prone empirical bridge obligations rather than as an already validated neuroscience theory.

Keywords: consciousness; panexperientialism; process ontology; causal structure; phenomenal structure; multiscale processes; subject individuation; evolution; self-model; falsifiability

1. The target problem

Contemporary consciousness theories assign central roles to integrated causal structure, recurrent processing, global availability, higher-order representation, predictive modeling, information closure, temporal organization, and related mechanisms (Seth & Bayne, 2022; Albantakis et al., 2023; Chang et al., 2020; Kanai & Fujisawa, 2024). Much of the competition among theories concerns whether one of these mechanisms constitutes, enables, or marks consciousness.

UCT begins by separating two questions:

Does experience exist?

and:

What is the organization of that experience?

UCT fixes the first question axiomatically on its declared domain of actual process tokens. It therefore relocates the nontrivial universal problem to experiential structure.

This move does not prove universal experience. It is an ontological wager. The scientific burden is correspondingly high: a theory that says every actual process has some experience cannot earn empirical value from that binary assertion alone. It must constrain structured differences, multiscale relations, transformation laws, and empirically testable bridge predictions.

The principal clarification of v1.1 is that the identity claim concerns the actual physical organization of a process token, not a representation chosen by an analyst. The second

is that finite experiments do not directly observe complete ontic organization or complete phenomenology. Operational tests therefore target explicitly declared finite bridge hypotheses.

2. Ontological domain: actual process tokens

2.1 Actuality and tokenhood

Let:

$$\mathcal{P}$$

be the domain of actual valid physical process tokens. An element:

$$P \in \mathcal{P}$$

is an actual causally continuous physical subhistory, not an arbitrary set of variables selected after the fact.

A valid token must satisfy, at minimum:

1. **actual support** — the relevant states/events are physically instantiated;
2. **induced-relation fidelity** — attributed internal relations are inherited from physical organization rather than invented by the modeller;
3. **causal continuity** — the token has a physically traceable causal history over the interval of interest;
4. **boundary accountability** — materially relevant cross-boundary influences are represented rather than silently erased;
5. **lineage traceability** — continuation, replacement, branching, merger, and cessation are physically trackable;
6. **description invariance** — invertible coordinate or label changes do not create new tokens.

UCT is process-token pluralist. Actual tokens may:

- overlap;
- nest;
- branch;
- merge;
- persist through replacement;
- exist at multiple physical scales.

No maximality or exclusion rule is imposed at the basal level.

2.2 Multiscale token plurality

A critical distinction in v1.1 is:

actual macroprocess token $\neq$ arbitrary coarse description.

Molecular, cellular, circuit-level, neural-system, organism-level, and other macroprocesses may all be actual process tokens when the relevant organization is physically instantiated. A scientist does not create a macroprocess merely by averaging microscopic variables.

This leaves a genuine implementation problem: what physical criteria are sufficient to identify an actual macroprocess token rather than a convenient model? UCT constrains the answer by requiring realized support, constitutive causal organization, reproducible

physical relations, and independence from the desired consciousness label. A fully general implementation theorem remains open.

2.3 Token, type, and lineage

UCT distinguishes:

$$\boxed{\text{structural type} \neq \text{numerical token} \neq \text{causal lineage.}}$$

Two process tokens can have isomorphic organization without being numerically identical:

$$\mathbf{D}(P) \cong \mathbf{D}(Q) \not\Rightarrow P = Q.$$

Lineage is a directed continuation relation rather than numerical identity. A process may branch:

$$P \rightsquigarrow Q_1, \quad P \rightsquigarrow Q_2, \quad Q_1 \neq Q_2,$$

or contribute to a merger:

$$P \rightsquigarrow M, \quad Q \rightsquigarrow M.$$

Nothing in these relations alone determines a unique subject count.

3. Ontic organization, views, estimates, and subjects

The central formal firewall of v1.1 is:

$$\boxed{\text{ontic process token} \neq \text{physical view} \neq \text{scientific estimate} \neq \text{subject.}}$$

3.1 Full token-relative ontic organization

For an actual process token P , define:

$$\boxed{\mathbf{D}_t^{\text{ontic}}(P)}$$

as the complete currently instantiated physical organization constitutive of that token.

“Complete” is token-relative. It means that every physical distinction or relation constitutive of P ’s own organization is included. It does not mean that every microscopic fact anywhere in the realizing substrate or surrounding universe is part of the same token.

This distinction permits fine and coarse actual process tokens to coexist without making every microscopic fluctuation a mandatory difference in a human-scale macroexperience.

3.2 Physical scientific views

Science does not ordinarily access $\mathbf{D}^{ontic}(P)$ directly. Instead, it constructs physically anchored views indexed by:

$$v = (t, \tau, g, \mathcal{I}, \mathcal{B}, \dots),$$

where t is time, τ a horizon, g a measurement/representation grain, $\mathcal{I}$ a physically admissible intervention family, and $\mathcal{B}$ a declared boundary representation.

Write:

$$D_v(P) = \Pi_v[\mathbf{D}^{ontic}(P)].$$

Different views need not form one total hierarchy. Some are refinements or coarse projections of others; some are incomparable.

Where two routes claim to construct the same target physical view, local consistency requires approximate or exact commutation:

$$R_{v \rightarrow z} \cong R_{w \rightarrow z} \circ R_{v \rightarrow w}.$$

3.3 Scientific estimates

Actual data yield:

$$\widehat{D}_v(P \mid \mathcal{E}, \mathcal{M}),$$

an estimate conditioned on evidence $\mathcal{E}$ and model class $\mathcal{M}$.

If evidence underdetermines the physical organization, the appropriate scientific output is an identified set:

$$\mathfrak{I}(P) = \{D : D \text{ remains compatible with the available evidence}\},$$

rather than an ontological point estimate chosen for convenience.

3.4 Subjecthood

A subject is not defined merely by being a process token, a scientific view, or a macroprocess.

UCT therefore maintains:

experience-bearing process $\neq$ canonically individuated subject.

Subject individuation is a separate theoretical problem.

4. Foundational commitments

A1 — Universal Experience

For every actual valid process token:

$$\boxed{\forall P \in \mathcal{P}, \quad \Phi(P) \neq \emptyset.}$$

A1 is not derived from evolution, neuroscience, intelligence, report, complexity, or biology.

Within some finite formalizations, A5 plus a nonempty process structure may make A1 mathematically redundant. It is retained explicitly because it states the intended ontological domain and prevents later definitions of “actual process” from quietly excluding simple cases.

A2 — Scoped Structural Continuity

For an explicitly declared continuously comparable family $P(\lambda)$, if the relevant physical organization varies continuously in a specified structural topology, then experiential organization varies continuously:

$$\boxed{\mathbf{D}(P_\lambda) \text{ continuous} \Rightarrow \Phi(P_\lambda) \text{ continuous.}}$$

A2 is a schema, not a universal scalar continuity law. Quantitative use requires a declared metric/topology. A derived categorical observable can undergo a threshold transition even when underlying structure varies continuously.

A3 — Persistence Under Embedding

If an actual process P remains physically realized while embedded in a larger actual process Q , embedding alone does not terminate P :

$$\boxed{P \triangleleft Q \not\Rightarrow \text{cessation of } P.}$$

The distinctive content of A3 is process ontological. Once the lower token remains actual, A1 supplies nonempty lower-scale experience.

A3 does not imply that a process continues after its constitutive dynamics cease.

A4 — Selfhood Non-Prerequisite

Human-like conceptual selfhood is not required for basal experience:

$$\boxed{\neg \text{Self}_I(P) \not\Rightarrow \Phi(P) = \emptyset.}$$

The positive mechanisms by which body models, agency, autobiographical memory, metacognition, and a conceptual “I” form are not axioms. They belong to higher-order application/bridge layers.

A5 — Experience-Structure Identity

Version 1.0 contained two components:

A5-W — one-way phenomenal completeness

$$\boxed{\mathbf{D}(P) \cong \mathbf{D}(Q) \Rightarrow \Phi(P) \cong \Phi(Q).}$$

This blocks a hidden phenomenal variable from changing while complete physical organization remains fixed.

A5-OI — ontic-token injective structural identity

Version 1.1 proposes the stronger, explicit development-branch commitment:

$$\boxed{\Phi(P) \cong \Phi(Q) \iff \mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q)}$$

for actual process tokens compared within a declared common structural signature.

A5-OI is stronger than the explicitly one-way v1.0 A5b formula. It is therefore a versioned strengthening/clarification, not a claim that v1.0 had already proved the converse.

It is weaker than an unrestricted global bijection between every logically possible phenomenal and physical structure.

A5 remains a metaphysical identity postulate. UCT does not derive it from nonphenomenal premises.

A6 — Token / Type / Lineage Distinction

$$\boxed{\mathbf{D}(P) \cong \mathbf{D}(Q) \nRightarrow P = Q.}$$

Structural equivalence does not imply numerical identity.

5. Formal consequences

5.1 Organization-Gate Impossibility

Define an experience-existence predicate:

$$E(P) = \mathbf{1}[\Phi(P) \neq \emptyset].$$

By A1:

$$E(P) = 1 \quad \forall P \in \mathcal{P}.$$

Let $F(P)$ be any organization descriptor and S a threshold/target region. If:

$$\exists Q \in \mathcal{P} : F(Q) \notin S,$$

then:

$$\boxed{\mathbf{1}[F(P) \in S] \neq E(P).}$$

Hence, within UCT, no nonuniversal threshold of:

- integration;
- recurrence;

- global availability;
- explicit memory;
- higher-order representation;
- predictive modeling;
- autonomy;
- entropy;
- generic biological complexity;

can equal the universal experience-existence predicate.

This theorem does not make those mechanisms irrelevant. It relocates them from experience existence to experience organization.

5.2 No Hidden Quale

From A5-W:

$$\mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q) \Rightarrow \Phi(P) \cong \Phi(Q).$$

UCT therefore rejects an additional hidden phenomenal degree of freedom that varies independently while complete token-relative physical organization is fixed.

Words such as “red” and “pain” are calibrated labels on experiential organization, not primitive unstructured quale atoms.

5.3 Probe Invariance

Let $\mathcal{C}^{ontic}(P)$ denote physically instantiated causal-dispositional organization and $\mathcal{J}^{exp}$ an experimenter-selected probe family.

The probe samples the process; it does not constitute the process.

Thus:

$$\mathcal{J}_1^{exp} \neq \mathcal{J}_2^{exp} \wedge \mathcal{C}_1^{ontic}(P) \cong \mathcal{C}_2^{ontic}(P) \Rightarrow \Phi_1(P) \cong \Phi_2(P).$$

What the scientist chooses to ask cannot, by itself, change what experience exists.

5.4 No Future Contamination

If:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q),$$

then a future external divergence that is not physically instantiated at time t cannot imply a present phenomenal difference:

$$\boxed{\Phi_t(P) \cong \Phi_t(Q).}$$

Present causal dispositions are part of the present organization; actual future facts are not.

5.5 Causally Screened History / No Ghost History

Suppose:

$$History(P) \neq History(Q)$$

but all current constitutive physical consequences of those histories have converged:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q).$$

Then:

$$\boxed{\Phi_t(P) \cong \Phi_t(Q).}$$

A6 still permits the tokens to remain numerically distinct because lineage can differ even when current structural type is the same.

5.6 External reward-label non-identity

Changing an external reward sign, semantic label, or experimenter convention without changing intrinsic constitutive organization cannot by itself alter phenomenology under A5-W.

Therefore:

$$\boxed{\text{reward sign} \neq \text{phenomenal valence sign.}}$$

6. Factorization, macroprocesses, and combination

6.1 Independent product

For physically independent process tokens P and Q , define an exact structural product:

$$\mathbf{D}(P) \otimes \mathbf{D}(Q)$$

whose joint transition/response organization factorizes across the two components.

This is a structural comparison object. It does not imply that every arbitrary set-theoretic pair constitutes a natural macroprocess token.

6.2 Actual macroprocess requirement

Before asking whether a macroprocess has nonproduct structure, M must first be independently identified as an actual process token.

A coarse description does not create M .

This is a major constraint on the combination discussion.

6.3 Non-Summative Combination — conditional on A5-OI

Let:

$$P_1, \dots, P_n \triangleleft M$$

be constituent actual tokens within an independently identified actual macroprocess M .

If:

$$\mathbf{D}^{ontic}(M) \not\cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

then under A5-OI:

$$\boxed{\Phi(M) \not\cong \bigotimes_i \Phi(P_i).}$$

Proof sketch. By A5-OI, the experiential structural type of each actual token is faithful to its complete token-relative ontic structural type. Assume for contradiction that:

$$\Phi(M) \cong \bigotimes_i \Phi(P_i).$$

Faithful structural identity would then require:

$$\mathbf{D}^{ontic}(M) \cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

contradicting the premise. The result is therefore conditional on A5-OI and on the independently established actual tokenhood of M .

This result does not say that micro-experiences fuse as substances. It says that when the actual whole process has constitutive relations not representable as independent juxtaposition, the whole experiential structure is likewise nonproduct.

A3 simultaneously permits constituent process experiences to continue when the constituent tokens continue.

Thus UCT replaces “How do microqualia combine into one substance?” with two distinct questions:

1. does a new actual macroprocess exist?
2. is its organization product or nonproduct relative to its constituents?

Subject count is not yet determined.

6.4 Macro invariance with micro change

Suppose two macroprocess tokens satisfy:

$$\mathbf{D}^{ontic}(M_A) \cong \mathbf{D}^{ontic}(M_B),$$

while some realizing lower-scale tokens differ:

$$\exists i : \mathbf{D}^{ontic}(p_i^A) \not\cong \mathbf{D}^{ontic}(p_i^B).$$

Then A5-OI permits:

$$\boxed{\Phi(M_A) \cong \Phi(M_B)}$$

while:

$$\Phi(p_i^A) \not\cong \Phi(p_i^B).$$

This makes substrate replacement, molecular turnover, and microphysical variation compatible with preservation of a macro experiential type, without declaring lower-scale differences phenomenally nonexistent.

7. Subjecthood without a free canonical partition

UCT distinguishes:

$$\text{experience} \neq \text{unity} \neq \text{macroprocess} \neq \text{subject} \neq \text{self-model}.$$

7.1 Symmetric-overlap construction

Consider three physically equivalent subsystems:

$$a, b, c$$

with reflection symmetry:

$$\rho : a \leftrightarrow c, \quad b \mapsto b.$$

Let:

$$P = \{a, b\}, \quad Q = \{b, c\},$$

with:

$$P \cap Q = \{b\} \neq \emptyset$$

and:

$$\mathbf{D}(P) \cong \mathbf{D}(Q).$$

Suppose a proposed subject selector S is required to depend only on the intrinsic physical/UCT structure and to preserve structural symmetry:

$$S(\rho W) = \rho S(W).$$

But:

$$\rho W = W.$$

Therefore:

$$S(W) = \rho S(W).$$

If S selects P but not Q , symmetry requires the converse. Contradiction.

Hence a structure-invariant selector must:

- select both;
- select neither;
- or introduce additional symmetry-breaking structure.

If the output is required to be one mutually exclusive subject partition, selecting both is impossible because the candidates overlap.

Therefore:

intrinsic structure alone does not always determine one unique mutually exclusive subject partition.

7.2 Scope

This theorem does not imply:

- subjects are unreal;
- no system ever has a unique subject;
- all subject partitions are arbitrary.

It applies to a specified class of symmetric, overlapping candidates under a structure-invariant mutually exclusive selector.

The result is a no-free-selection theorem, not a universal theory of subjecthood.

8. From physical process to conceptual self: a structural-evolutionary synthesis

This section is explanatory rather than foundational.

It asks:

If A1 and A5-OI are granted, can one describe a continuous and branching path from elementary physical processes to human conceptual selfhood without introducing successive new phenomenal substances?

UCT can represent such a path coherently as a structural synthesis. This does not establish that the historical transitions occurred in exactly the sequence listed below, nor does it provide independent evidence for A1 or A5-OI. The relevant theoretical claim is narrower: no additional phenomenal substance is required by the UCT formalism when these organizational transformations are described.

Crucially:

evolutionary continuity $\nRightarrow$ A1

and:

evolutionary continuity $\nRightarrow$ A5.

An emergentist may accept Darwinian continuity while positing an experience threshold. The structural-evolutionary synthesis therefore tests explanatory coherence; it does not prove the axioms.

8.1 Elementary physical processes

At the most elementary level considered by UCT, an actual interaction process possesses whatever relational structure is physically instantiated.

UCT does not require:

- a nervous system;
- a sensor;
- a memory;
- a self-model;
- a report channel.

Under A1, the experience-existence predicate is already nonzero. Under A5, the experiential structure is identified with the actual process organization.

UCT therefore refuses to project human terms such as “seeing,” “wanting,” or “suffering” onto elementary interactions. The primitive claim is structural, not anthropomorphic.

8.2 From stable chemistry to self-maintaining organization

The transition from nonliving chemistry to the first cells is an origin-of-life problem, not a solved Darwinian lineage. UCT does not claim to reconstruct the historical path.

The theoretical point is narrower. If physical processes develop persistent boundaries, recurrent chemical organization, and endogenous regulation, then new relations appear inside the process organization:

internal state $\leftrightarrow$ boundary $\leftrightarrow$ environmental perturbation $\leftrightarrow$ regulatory response.

These relations are not the origin of experience under A1. They are transformations of experiential organization.

8.3 Single-cell and non-neural learning: temporal depth

A decisive organizational transition occurs when past events leave present physical traces that modify current response:

$$History \rightarrow m_t \rightarrow Response_t.$$

This is the minimal form of temporal depth required by UCT’s memory analysis.

The framework does not require memory to create experience. Memory changes current organization.

Non-neural learning illustrates the physical possibility of history-dependent behavior without a brain. Habituation has been experimentally demonstrated in the slime mould *Physarum polycephalum* (Boisseau, Vogel, & Dussutour, 2016). Such evidence does not establish phenomenology in the organism; it demonstrates that learning-like temporal organization need not await centralized nervous systems.

If all physical consequences of a history are erased, No Ghost History predicts no remaining current phenomenal-type difference attributable solely to the past.

8.4 Multicellularity: from juxtaposition to organism-scale relations

Consider independent cells:

$$C_1, \dots, C_n.$$

If their joint organization remains a product, there is no new irreducible cross-cell relation merely because the cells are spatially adjacent.

If, however, real chemical, electrical, mechanical, developmental, or other causal relations generate an actual multicellular process M with:

$$\mathbf{D}(M) \not\cong \bigotimes_i \mathbf{D}(C_i),$$

then A5-OI assigns M a nonproduct experiential structure.

The significance of multicellularity in UCT is therefore not cell count. It is the formation of new actual macroprocess organization.

No privileged signaling medium is required. Mechanical, chemical, electrical, and neural relations can all contribute when they are physically constitutive.

8.5 Nervous systems: routing and specialization, not an experience switch

The evolution of neurons adds high-speed, long-range, selectively routed, and plastic coordination.

Under UCT:

$\text{neuron} \neq \text{experience ignition.}$

Neurons transform organization. They can increase spatial discrimination, routing, temporal coordination, learning capacity, and context sensitivity.

Associative learning in the box jellyfish *Tripedalia cystophora*, a cnidarian without a vertebrate-style centralized brain, provides an empirical illustration that nontrivial learning need not wait for vertebrate-like centralization (Bielecki et al., 2023).

This finding is not evidence for UCT's A1. It illustrates the danger of equating a particular level of centralized neural architecture with the first appearance of complex learning.

8.6 Damage regulation, nociception, and affect

The pain lineage is particularly useful because it exposes the mistake of projecting a fully human quale backward.

Comparative biology indicates that damage detection and adaptive cellular stress responses predate nervous systems, while specialized nociceptors, sensitization, central aversive states, and human pain involve progressively richer mechanisms (Walters, 2025).

A schematic structural trajectory is:

cellular damage regulation $\rightarrow$ multicellular injury coordination $\rightarrow$ specialized nociception $\rightarrow$ sensitization

UCT does not identify the first term with human pain.

It instead asks which physical relations constituting later pain are already present in simpler systems.

Human pain also illustrates why one scalar valence variable is inadequate. Clinical and conceptual literatures distinguish sensory-discriminative from affective-motivational dimensions of pain, and pain-asymbolia cases have been interpreted as evidence for partial dissociation between them (Auvray, Myin, & Spence, 2010). That interpretation is not uncontested: recent philosophical reviews emphasize that asymbolia may reflect strongly

reduced rather than literally absent negative affect, and that the taxonomy of the residual state remains debated (Coninx, 2024). The conservative UCT conclusion is therefore:

nociceptive discrimination, unpleasantness, and suffering
are not licensed as one primitive scalar variable.

The exact physical organization grounding human unpleasantness remains open.

8.7 Association, relational abstraction, and reasoning

Cognitive evolution can likewise be represented as transformations of organization rather than the sudden arrival of a new phenomenal substance.

A simplified comparative sequence is:

state-dependent response → habituation / sensitization
→ association → relational abstraction
→ relations over relations
→ symbolic / compositional reasoning.

The sequence is not a literal phylogenetic ladder.

Honeybees can learn and transfer same/different relational rules to novel stimuli (Giurfa et al., 2001), supporting the narrower claim that transferable relational discrimination need not require a human-sized brain.

Cuttlefish have shown what-where-when behavioral organization meeting standard criteria for episodic-like memory (Jozet-Alves, Bertin, & Clayton, 2013), and later work demonstrated substantial delayed control in a delay-of-gratification task (Schnell et al., 2021).

These examples support the narrower conclusion that component capacities relevant to human cognition occur in distant lineages and can be realized by substantially different nervous-system architectures. They do not establish a single linear ancestry of cognition, nor do they demonstrate that nonhuman animals execute human formal logic or possess phenomenology identical to human cognition.

8.8 Physical witness and semantic inflation

A danger in comparative cognition is to redescribe simple behavior with sophisticated human concepts.

UCT therefore introduces a Physical Witness Criterion:

A claimed higher-order relation belongs to the actual constitutive organization only if the physical process itself distinguishes that relational configuration under controlled, physically admissible perturbation.

For example, “same/different” should not be attributed merely because an analyst can describe stimuli relationally. Transfer to novel physical stimuli provides stronger evidence that the system is sensitive to an invariant relation, though behavioral transfer alone still need not reveal the complete ontic mechanism.

The criterion blocks:

observer’s description ⇒ invented phenomenal dimension.

8.9 Body-centered organization

With increasingly coordinated sensorimotor organization, a system can physically encode relations involving:

- internal/external boundary;
- body position;
- sensor position;
- action state;
- consequences of self-generated action;
- persistent internal variables.

Write this only as a projection of already existing process organization:

$$\mathcal{S}(M) = \Pi_{sys}[\mathbf{D}^{ontic}(M)].$$

This is system-relative organization, not a new observer inside the system.

It can support increasingly rich body-centered and agency-related experience without being the origin of experience itself.

8.10 Self-model

A self-model is a high-level physical organization in which some internal states track, predict, compress, or regulate aspects of the system's own organization across time. Possible ingredients include:

- body state;
- action consequences;
- memory;
- prediction;
- social information;
- control;
- metacognitive access.

UCT does not define the self-model as the subject, and it does not make self-modeling necessary for basal experience.

8.11 The conceptual "I"

Human conceptual selfhood adds language, autobiographical structure, explicit temporal narration, symbolic categorization, social cognition, metacognition, and culturally transmitted conceptual tools.

A system can then instantiate structures corresponding to propositions such as:

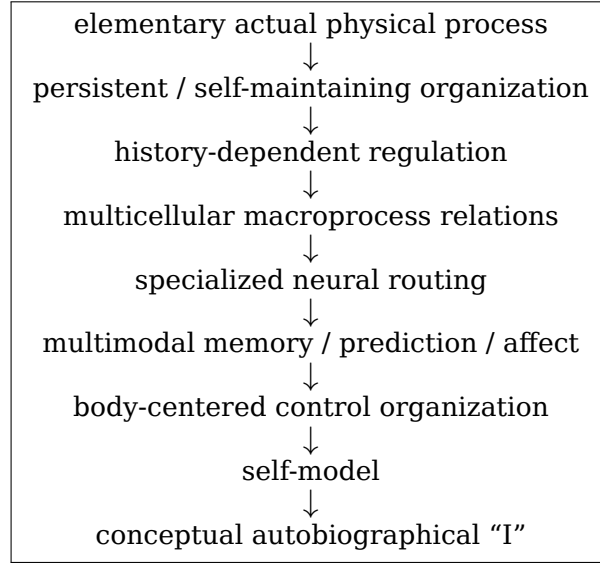
I am the one undergoing this experience.

or:

I remember being the same individual who experienced that event.

Within UCT, this conceptual "I" is a late, high-dimensional organization inside experience, not the entity that creates experience.

The proposed structural sequence is therefore:



The arrows mean structural transformation, not one linear species ancestry.

The comparative cases in this section are therefore evidential anchors for organizational possibilities and intermediate capacities. They are not empirical confirmation of universal experience, structural identity, or cross-species quale identity.

8.12 No universal consciousness ladder

Evolution can:

- add relations;
- delete relations;
- reweight couplings;
- prune structure;
- specialize;
- compress irrelevant details;
- introduce new temporal dependencies;
- bifurcate dynamically;
- lose capacities;
- converge on similar functions by different architectures.

Therefore UCT rejects:

$$\frac{dC}{dt} > 0$$

for a universal scalar amount of consciousness.

The more general representation is:

$$\lambda \mapsto \mathbf{D}_{\lambda}^{ontic}(P_{\lambda}) \mapsto \Phi_{\lambda}(P_{\lambda}).$$

Evolution is a path through a branching space of experiential organization, not a staircase from unconscious matter to conscious humans.

9. Structural transformation versus enrichment

A previous development branch emphasized relational enrichment:

$$D_1 \prec_E D_2.$$

Strict enrichment remains useful when D_2 preserves the distinctions of D_1 while adding new, nonredundant constitutive relations.

But biological change is more general.

Let:

$$T : \mathbf{D}_1 \mapsto \mathbf{D}_2.$$

Possible transformations include:

- current-state change;
- coupling reweighting;
- relation addition;
- relation deletion;
- pruning;
- factorization or refactorization;
- temporal-depth change;
- abstraction/compression;
- dynamic bifurcation;
- specialization.

Thus:

$\text{enrichment} \subset \text{structural transformation.}$

Under A5-OI:

$$\mathbf{D}_1 \not\cong \mathbf{D}_2 \Rightarrow \Phi_1 \not\cong \Phi_2.$$

This is a direct consequence of the identity commitment, not an independent discovery or an additional empirical law.

10. Scientific bridge architecture

A5-OI concerns complete ontic organization and complete experiential structure.

Ordinary science observes neither completely.

Finite empirical tests therefore require explicit auxiliary structures.

10.1 Declared phenomenal target

Let:

$$\Psi_v(P)$$

be a finite, preregistered phenomenal/psychophysical target inferred from some combination of:

- similarity judgments;

- discrimination performance;
- subjective ratings;
- confidence;
- no-report perceptual indicators;
- multidimensional report patterns;
- physiological measures with independently justified interpretation.

By default:

$$\Psi_v(P) \neq \Phi(P).$$

It is a scientific projection, not complete phenomenology.

10.2 Bridge Sufficiency Hypothesis

Define:

$$B_v : D_v(P) \rightsquigarrow \Psi_v(P)$$

as the hypothesis that the frozen physical view retains the distinctions sufficient for the declared phenomenal target.

B_v is not A5.

It can fail.

10.3 Operational Structural Correspondence

The testable package is:

$$OSC_v = A5 + B_v + \text{measurement assumptions.}$$

A one-way version predicts:

$$D_v(A) \cong D_v(B) \Rightarrow \Psi_v(A) \cong \Psi_v(B).$$

An injective version predicts:

$$D_v(A) \cong D_v(B) \iff \Psi_v(A) \cong \Psi_v(B)$$

within the preregistered bridge scope.

A finite empirical failure directly challenges the operational package. It does not automatically isolate A5 from the bridge and measurement assumptions.

11. Physical Token/View Selection Protocol

The main falsifiability danger for UCT is post-hoc redefinition.

If a failed experiment can always be answered by choosing a new token, scale, horizon, intervention family, or structural descriptor after seeing the phenomenal result, the theory becomes scientifically non-vulnerable.

Version 1.1 therefore proposes the Physical Token/View Selection Protocol (PTVSP).

Before phenomenal unblinding:

1. declare candidate actual-token hypotheses;
2. declare admissible physical views;
3. declare physical intervention families;
4. declare held-out physical response targets;
5. evaluate physical predictive sufficiency;
6. evaluate stability under measurement perturbation;
7. evaluate cross-system reproducibility when possible;
8. prefer parsimonious physically sufficient views or a preregistered Pareto set;
9. freeze token/view/bridge definitions;
10. only then join or reveal phenomenal outcomes.

The central rule is:

No Post-Hoc Token/View Rescue.

A failed preregistered bridge remains failed.

Changing token, view, grain, horizon, or bridge afterwards creates a new version of the hypothesis.

12. Empirical failure conditions

12.1 Same frozen physical view, different phenomenal target

If:

$$D_v(A) \cong D_v(B)$$

but:

$$\Psi_v(A) \not\cong \Psi_v(B),$$

then the preregistered one-way operational bridge fails.

If B_v and measurement completeness were independently secure, this would pressure A5-W directly. In ordinary practice, it falsifies the joint package first.

12.2 Different frozen physical views, identical phenomenal target

Under an injective operational bridge, if:

$$D_v(A) \not\cong D_v(B)$$

but:

$$\Psi_v(A) \cong \Psi_v(B),$$

then the injective operational bridge fails.

Again, attribution specifically to A5-OI requires control of auxiliary assumptions.

12.3 Behaviorally matched causal rewiring

A particularly informative design would match:

- overt behavior;
- report;
- task accuracy;
- confidence;
- access variables;

while manipulating a preregistered constitutive causal relation that survives the frozen physical view.

UCT's A5-OI bridge predicts a difference in complete experiential organization even when the chosen output variables match.

This places UCT within the empirical pressure created by unfolding/substitution arguments against causal-structure theories (Doerig et al., 2019; Kleiner, 2020; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri, BarAvi, & Hemmo, 2026).

12.4 Latent-memory test

Match explicit current report and task content while preserving a preregistered difference in latent constitutive memory dispositions that changes future admissible response.

A5-OI predicts a difference in complete experiential organization, though not necessarily a difference in explicitly recalled content.

This is a risky UCT-specific consequence once the relevant physical view and bridge are frozen.

13. Relation to UCT II and neighboring theories

UCT II asks whether operational cores of major consciousness theories can be reconstructed, embedded, reinterpreted, or shown to conflict as effective organization theories on top of the UCT mother formalism.

The relation is asymmetric:

failure of one UCT-II bridge $\nRightarrow$ failure of the basal ontology

but:

failure of A1/A5 $\Rightarrow$ revision of the UCT-specific interpretation of UCT II.

UCT II introduced explicit Bridge Admissibility Conditions, Translation Fidelity Requirements, formal recovery classes, and separate empirical-validation levels. Those constraints remain essential for preventing a rich mother formalism from trivially “containing” other theories by lookup.

Version 1.1 does not claim novelty for:

- panexperientialism or panpsychism;
- process-oriented metaphysics of experience;
- mind-brain or mind-physical identity strategies in general;
- causal/phenomenal structuralism;
- multiscale process accounts;
- overlapping conscious systems;
- mechanistic physical implementation criteria;

- structural approaches to subjectivity;
- the general idea that theories may be partially complementary.

Several neighboring traditions overlap with individual UCT commitments. Panexperientialist and process traditions already treat experience as widespread or fundamental rather than as a late biological product (e.g., Rosenberg, 2004, building on earlier process traditions). IIT explicitly advances an explanatory identity between phenomenal structure and a substrate's intrinsic cause-effect structure (Albantakis et al., 2023), while neurophenomenal structuralism develops a relational/structural account of specific phenomenal character (Lyre, 2022). General mind-brain identity theories likewise deny that mental and physical processes are merely correlated. UCT therefore does not claim to originate the idea of identity itself.

The combination, de-combination, overlap, and individuation problems are also established literatures (Chalmers, 2016; Miller, 2018; Blackmon, 2021). Recent preprint-level structural-subjectivity programs go further by proposing positive structural criteria for the presence or locus of subjectivity; for example, Nagae (2026) identifies subjectivity with viability-constrained irreversible exclusive commitment. UCT differs at this point: it does not use action, viability, irreversibility, recurrence, access, self-modeling, or any other nonuniversal organization as a basal experience-existence gate, and it presently offers only a scoped no-free-selection theorem rather than a universal positive subject criterion.

Neighboring and critical literatures also include Information Closure Theory (Chang et al., 2020), universal-theory criteria (Kanai & Fujisawa, 2024), mechanistic computation and implementation accounts (Piccinini, 2015; Anderson & Piccinini, 2024), and the unfolding/substitution debate (Doerig et al., 2019; Kleiner, 2020; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri et al., 2026).

The contribution claimed here is therefore not any single ingredient. It is the following constrained conjunction and dependency architecture:

universal process-token experience
 +ontic-token structural identity
 +nested non-exclusion
 +token/type/lineage separation
 +macroprocess nonproduct realization
 +experience/subject separation
 +no-free canonical subject extraction in symmetric overlap
 +explicit ontic/view/estimate firewall
 +failure-prone finite bridge methodology.

A targeted prior-art search did not identify an earlier source containing this exact conjunction with the same dependency architecture, but this is not a certified global priority claim; historical priority remains open.

14. Strong objections and limitations

14.1 The axioms are strong

A1 and A5-OI are not empirical discoveries.

They are strong starting commitments.

The formal theorems derived from them must therefore be evaluated partly by the explanatory costs and empirical obligations of the axioms.

14.2 A5-OI risks tautological explanatory rhetoric

Once structural identity is assumed, results of the form:

$$D \neq D' \Rightarrow \Phi \neq \Phi'$$

are straightforward consequences.

They should not be marketed as deep independent discoveries.

The nontrivial content must come from:

- process individuation;
- physical anchoring;
- exact no-go consequences;
- multiscale relations;
- subject non-selection;
- independently constrained bridge predictions.

14.3 Complete ontic organization is not directly observed

Science works with:

$$D_v$$

and:

$$\widehat{D}_v.$$

Direct empirical isolation of full A5-OI from bridge assumptions is therefore difficult.

UCT should not disguise this epistemic limitation.

14.4 Complete phenomenology is not directly measured

Human report, behavior, psychophysics, and physiology are projections.

Even convergent measures do not automatically equal complete phenomenal structure.

This limits the strength of empirical falsification claims.

14.5 Process proliferation

A permissive process ontology can yield many weak, overlapping, or nested experience-bearing tokens.

UCT treats this as an ontological cost, not a defect to be erased by an arbitrary threshold.

Scientific salience and experience existence are separate questions.

14.6 Macroprocess implementation remains incomplete

The statement that a macroprocess is an actual token must be grounded physically rather than phenomenologically.

General criteria for when a coarse organization counts as a genuine realized macroprocess remain incomplete.

14.7 Positive subject individuation remains open

UCT provides a scoped no-free-selection result.

It does not yet provide a universal positive law that identifies exactly one subject or subject count in every physical system.

14.8 Valence remains incomplete

UCT can represent physically directed regulatory organization and trace structural continuities from damage regulation to human pain.

But A1–A6 do not independently assign semantic positive/negative signs, and the exact physical organization grounding human unpleasantness remains open.

14.9 Cross-species qualia identity is underidentified

Similar behavior, homologous mechanisms, or structurally comparable organization do not by themselves prove identical subjective qualities across species.

Claims should therefore use terms such as:

- precursor;
- conserved mechanism;
- structural homology;
- convergent similarity;

rather than asserting uncalibrated qualitative identity.

14.10 Relativistic and quantum extensions

The present finite/classical formalism is not a final fundamental-physics theory.

A relativistic version should be formulated over causal spacetime histories rather than a preferred global time slice.

A quantum version requires appropriate state spaces, channels/instruments, subsystem criteria, and intervention notions.

14.11 Semantic representation

Causal tracking does not automatically establish semantic aboutness.

Naturalized representation, conceptual content, and explicit self-reference require additional theory.

15. Claim-status summary

Claim	Status in v1.1
Universal experience for actual valid process tokens	Axiom A1
Scoped continuity of experiential organization	Axiom-schema A2
Embedding does not terminate a continuing token	Axiom A3
Human-like conceptual self not required for basal experience	A4 boundary commitment
One-way phenomenal completeness	inherited A5-W

Claim	Status in v1.1
Ontic-token injective structural identity	proposed strengthened A5-OI
Token/type/lineage distinction	Axiom A6
Organization-Gate Impossibility	theorem conditional on A1 + nonuniversality witness
No Hidden Quale	consequence of A5-W
Probe Invariance	consequence of ontic/probe separation + A5-W
No Future Contamination	consequence of A5-W
Causally Screened History	consequence of A5-W + A6
Non-Summative Combination	conditional theorem requiring A5-OI + actual macro token
Macro invariance with micro change	conditional theorem requiring A5-OI
Symmetric-overlap non-selection	scoped structural theorem
Canonical positive subject selector	open
Valence sign from A1-A6	underdetermined
H-V* as basal valence ontology	rejected/demoted
RREP as universal generator	rejected/demoted
Structural evolution from physical process to conceptual self	explanatory/application synthesis
Evolutionary continuity entails A1/A5	explicitly rejected
PTVSP	methodological proposal
OSC _v	empirical bridge hypothesis
A1/A5 empirically established	not established
Hard problem deductively solved	not claimed
Subject-combination problem fully solved	not claimed
Individual ingredients are unprecedented	explicitly not claimed
Exact v1.1 conjunction found in targeted search	no earlier exact match located; not a global priority certification
Historical priority	not established

16. Conclusion

UCT I v1.1 retains the central wager of v1.0 while making its ontology and scientific obligations more explicit.

The basal claim is not that complexity creates consciousness. It is that actual physical processes possess experiential structure and that, under A5-OI, the structural type of that experience is identical to the token-relative ontic organization of the process.

This move changes the explanatory target.

The theory no longer asks:

At what threshold does nonexperience become experience?

It asks:

Which actual processes exist, what is their constitutive organization, how does that organization transform, and what empirical finite views preserve the distinctions relevant to declared phenomenal targets?

Within this framework, elementary interaction, self-maintenance, memory, multicellular coordination, neural routing, affect, relational cognition, self-modeling, and the human conceptual “I” can be placed on one structural map without adding a new phenomenal substance at each stage.

The evolutionary synthesis is not a proof of universal experience. It is a stress test of explanatory unification. Its value is that the same core does not need to switch ontologies when moving from simple physical organization to cells, animals, brains, and reflective humans.

The resulting theory is deliberately costly. It accepts process plurality, nested experience, incomplete subject individuation, difficult empirical identification, and a metaphysical identity commitment that cannot presently be isolated from all auxiliary bridge assumptions.

Those costs should not be hidden.

The scientific future of UCT therefore depends not on further expansion of the ontology, but on increasingly strict bridge tests in which process-token hypotheses, physical views, and mapping rules are fixed independently of phenomenal results and are allowed to fail.

The next stage of the program should therefore be empirical rather than metaphysical: pre-registered tests of finite Operational Structural Correspondence against named alternative theories.

References

Layout note for this release candidate. DOI identifiers are rendered as hyperlinks so long identifiers can break safely in the PDF without changing bibliographic content.

Reference-status note. Literature added specifically for the v1.1 evolutionary/comparative synthesis was checked against publisher, PubMed, or journal records during the 29 September 2026 claim audit. Basri et al. (2026) was indexed by PubMed as ahead-of-print on 8 September 2026 even though the journal issue is dated November 2026. The prior-art audit also added Rosenberg (2004) as an explicit panexperientialist comparison and Nagae (2026) as a recent preprint-level structural-subjectivity comparator.

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Appendix A. Main changes relative to v1.0

1. Replaces the potentially ambiguous scale-indexed ontological reading with a token-relative ontic organization:

$$\Phi(P) \leftrightarrow \mathbf{D}^{ontic}(P).$$

2. Separates:

$$\mathbf{D}^{ontic} \neq D_v \neq \widehat{D}_v \neq Subject.$$

3. Introduces the explicit A5-W / A5-OI distinction.
4. Marks all converse/nonisomorphism theorems as A5-OI-dependent.
5. Re-proves Non-Summative Combination with the actual-macro-token premise made explicit.
6. Adds Probe Invariance, No Future Contamination, and Causally Screened History.
7. Adds the scoped symmetric-overlap non-selection theorem.
8. Recasts A3 primarily as process persistence under embedding.
9. Recasts A4 as Selfhood Non-Prerequisite and removes positive self-formation mechanisms from the axiomatic layer.
10. Demotes First-Person Reference Field, binary macrofield membership, H-V*, and RREP-as-universal-generator from the foundational layer.
11. Adds the structural-evolutionary synthesis from elementary physical processes to conceptual selfhood.
12. Introduces the distinction:

$$A5 \neq B_v \neq OSC_v,$$

making finite empirical bridge assumptions explicit.

13. Replaces grain-only falsifiability discipline with Physical Token/View Selection Protocol and No Post-Hoc Token/View Rescue.
14. States explicitly that finite experiments ordinarily test an operational bridge package rather than isolated full A5-OI.
15. Expands the open-problem ledger to include macroprocess implementation, approximate structural metrics, direct A5 identification, complete phenomenology measurement, positive subject individuation, semantic representation, cross-species calibration, relativity, and quantum theory.